

MEDICAL Q&A CHATBOT

Professional Project Report

Project	Retrieval-Based Medical Question Answering
Dataset	MedQuAD
Interface	Streamlit Web Application
Core Retrieval	TF-IDF + Cosine Similarity
Entity Recognition	Basic Medical NER
Submitted By	Sujata Mali

Project purpose: Develop a specialized medical question-answering chatbot that retrieves relevant answers from the MedQuAD knowledge base and presents them through a simple, interactive interface.

Safety notice: The chatbot provides educational information retrieved from the dataset. It is not a substitute for professional medical advice, diagnosis, or treatment.

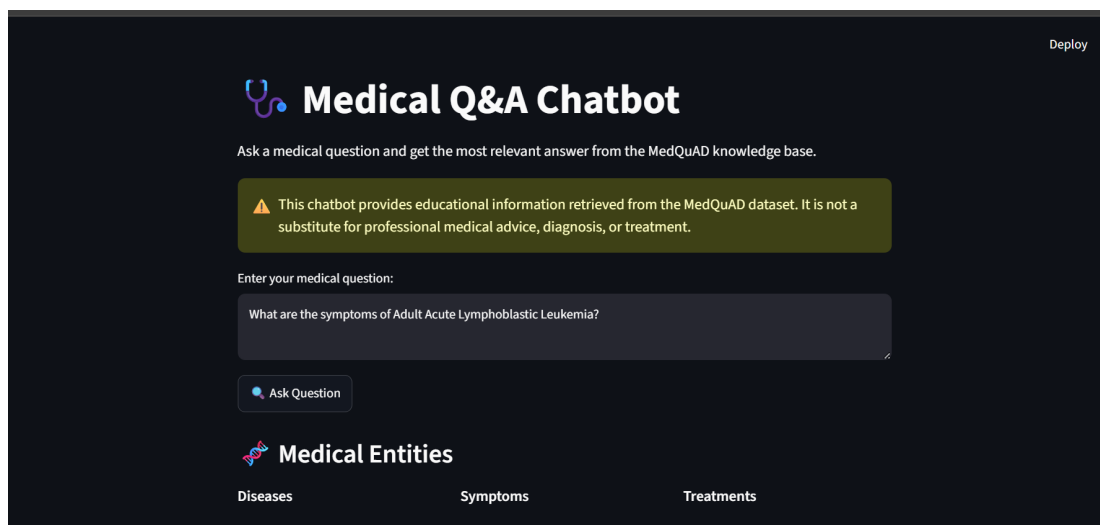


Figure 1. Streamlit chatbot home screen and medical question input interface.

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Repository

https://github.com/SujataMali03/Medical_QA_Chatbot

The repository contains the chatbot source code, project files, dataset-processing components, and deployment-related files.

How the screenshots are used in this report

The following application screenshots are included as project evidence. Each screenshot is placed directly below the relevant description so the professor can connect the implementation explanation with the working interface.

1. Project Overview

The Medical Q&A Chatbot is a domain-specific retrieval-based chatbot built using the MedQuAD medical question-answering dataset. Instead of generating medical advice, the application searches a curated medical knowledge base and returns the answer associated with the most relevant question. The application also displays retrieval information such as similarity, medical focus, matched question, question type, and basic medical entities. A relevance threshold helps prevent clearly unrelated questions from being answered as if they were medical questions.

2. Project Objectives

- Build a specialized medical question-answering chatbot using MedQuAD.
- Implement a retrieval mechanism to identify relevant medical answers.
- Use TF-IDF and cosine similarity for question matching.
- Perform basic medical entity recognition for diseases, symptoms, and treatments.

- Provide an easy-to-use Streamlit interface.
- Handle unsupported or non-medical questions safely.

3. MedQuAD Dataset

MedQuAD is a medical question-answering dataset containing question-answer pairs collected from health information resources. The project uses the downloaded MedQuAD repository, which contains multiple source folders such as CancerGov, GARD, GHR, MedlinePlus health topics, NIDDK, NINDS, SeniorHealth, NHLBI, CDC, and other medical sources.

The dataset is processed into a searchable knowledge base. Each question becomes a retrieval document, while its associated answer is returned when that question is identified as the closest match to the user's query.

4. Technology Stack

Technology	Purpose
Python	Core programming language and application logic.
MedQuAD	Medical question-answer knowledge base.
TF-IDF	Converts questions into numerical text vectors.
Cosine Similarity	Measures similarity between user question and dataset questions.
Basic Medical NER	Identifies diseases, symptoms, and treatments using rules/entity patterns.
Streamlit	Creates the interactive web-based chatbot interface.

5. System Architecture & Workflow

The application follows a retrieval pipeline: the user enters a medical question; the question is cleaned and vectorized; TF-IDF represents the query and dataset questions numerically; cosine similarity compares the query against the knowledge base; the highest-scoring relevant question is selected; its answer and metadata are displayed; and a threshold is used to reject weak or unrelated matches.

Step	Process
1	User enters a medical question.
2	Question preprocessing / normalization.
3	TF-IDF vectorization.
4	Cosine similarity calculation.
5	Best matching MedQuAD question is selected.
6	Medical entities and question type are displayed.
7	Answer is shown with safety notice.
8	Low-relevance / non-medical query is rejected.

6. Retrieval Mechanism

TF-IDF (Term Frequency-Inverse Document Frequency) represents each question according to the importance of its words. Common words receive less weight while words that are informative for a particular question receive more weight.

Cosine Similarity compares the vector of the user's question with vectors representing questions in the MedQuAD knowledge base. A higher similarity score means that the two questions are more closely related in the TF-IDF vector space.

Relevance threshold: If the best similarity score is below the configured threshold, the application does not present the result as a sufficiently relevant medical answer. This is important for preventing unrelated questions from being treated as valid medical queries.

7. Basic Medical Entity Recognition

The application provides basic medical entity recognition to highlight useful medical concepts in the user's question. The interface separates detected items into three groups:

- Diseases: medical conditions such as asthma, leukemia, diabetes, and other recognized conditions.
- Symptoms: symptom-related terms such as fever, fatigue, coughing, pain, and similar concepts.
- Treatments: treatment-related terms such as medicines, therapies, or treatment concepts recognized by the project.

This is a basic project-level NER feature rather than a clinical diagnostic system. Its purpose is to demonstrate entity extraction and make the chatbot's retrieval output easier to understand.

8. Streamlit User Interface

The Streamlit interface provides a simple web page where a user can enter a medical question and click Ask Question. The result area can display the detected medical entities, question type, similarity score, medical focus, matched question, and retrieved answer. The interface also shows a prominent educational-information warning.

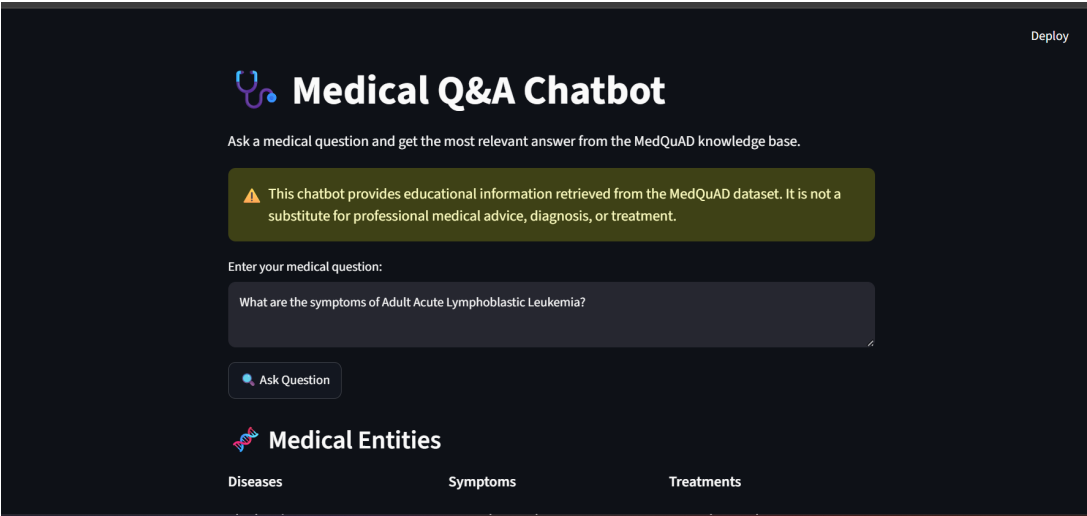


Figure 2. Main Streamlit interface showing the chatbot title, safety notice, input field, and medical entities area.

9. Testing & Results

The chatbot was tested with medical questions covering different conditions and question types. The following tests demonstrate successful retrieval as well as safe rejection of unrelated questions.

Test	Example	Observed behavior
Symptoms	What are the symptoms of Adult Acute Lymphoblastic Leukemia?	Relevant MedQuAD answer retrieved with high similarity.
Treatment	What are the treatments for leukemia?	Relevant treatment information retrieved.
Asthma	How is asthma diagnosed?	Relevant asthma answer retrieved.

Test	Example	Observed behavior
Diabetes	What are the symptoms of diabetes?	Relevant diabetes information retrieved.
Non-medical	How can I learn Python?	Rejected as insufficiently relevant to the medical knowledge base.
Non-medical	What is the capital of India?	Rejected as a non-medical question.

10. Screenshots / Application Evidence

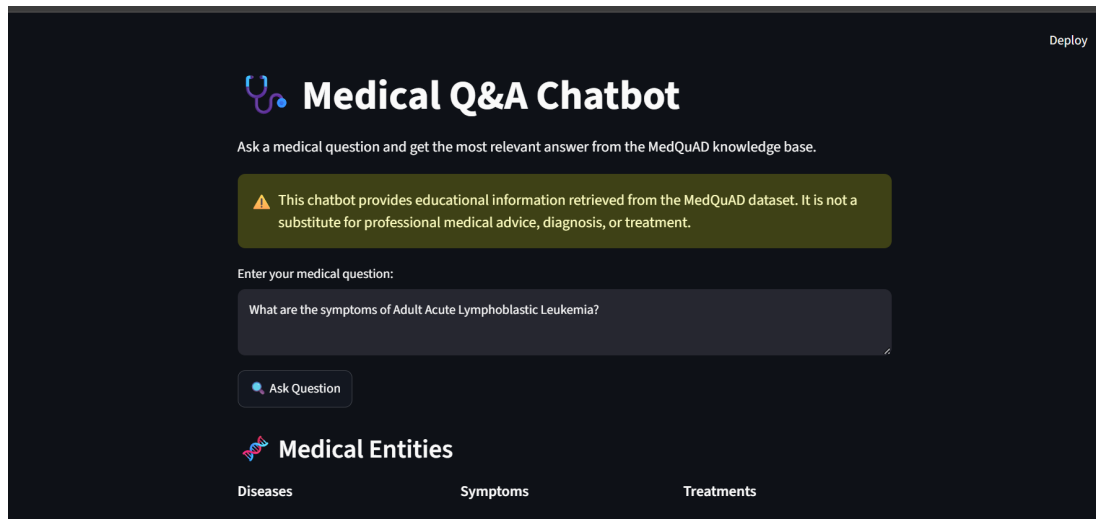


Figure 3. Application landing page with medical safety notice and question input.

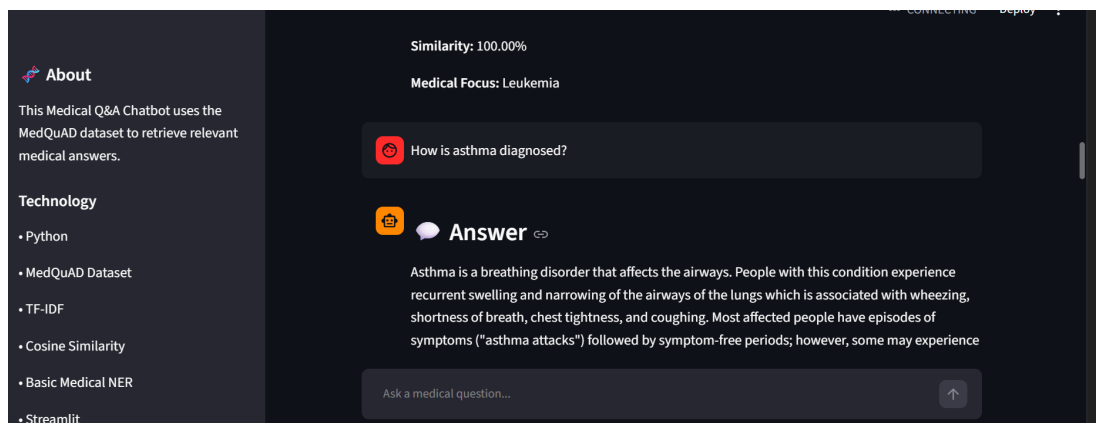


Figure 4. Asthma query and retrieved answer displayed with similarity information.

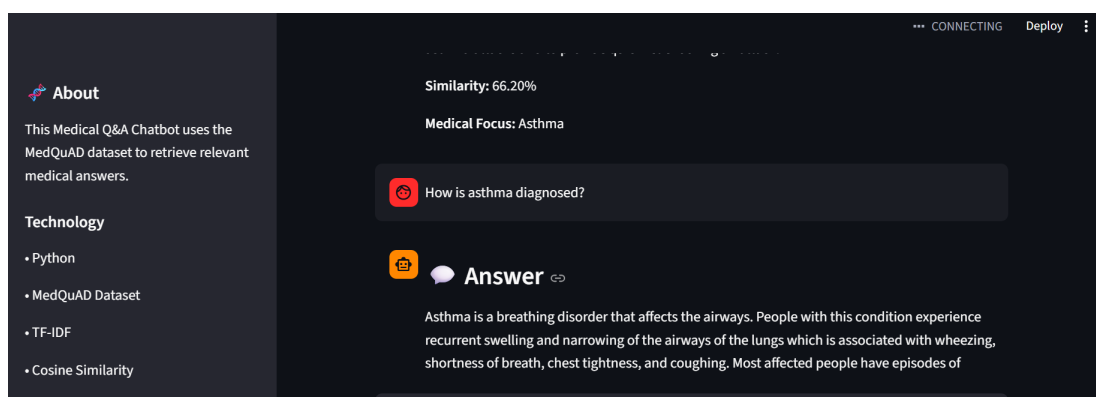


Figure 5. Asthma retrieval result showing the medical focus and answer.

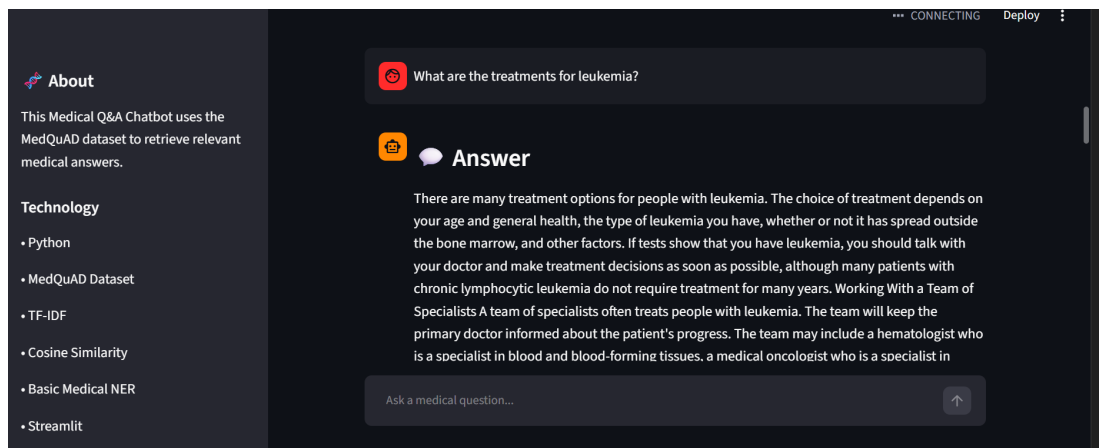


Figure 6. Leukemia treatment question and retrieved MedQuAD answer.

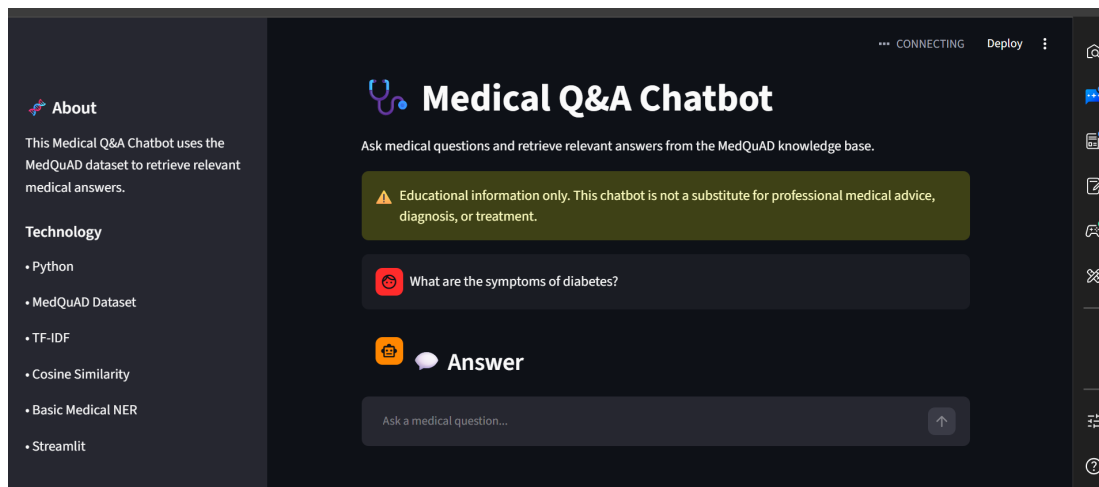


Figure 7. Diabetes symptoms query and retrieved answer in the Streamlit interface.

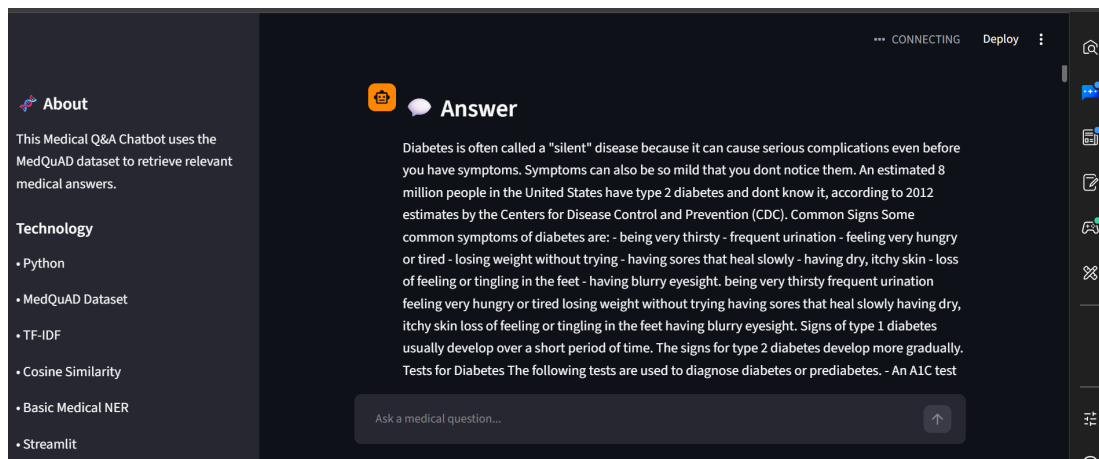


Figure 8. Detailed diabetes answer returned by the retrieval system.

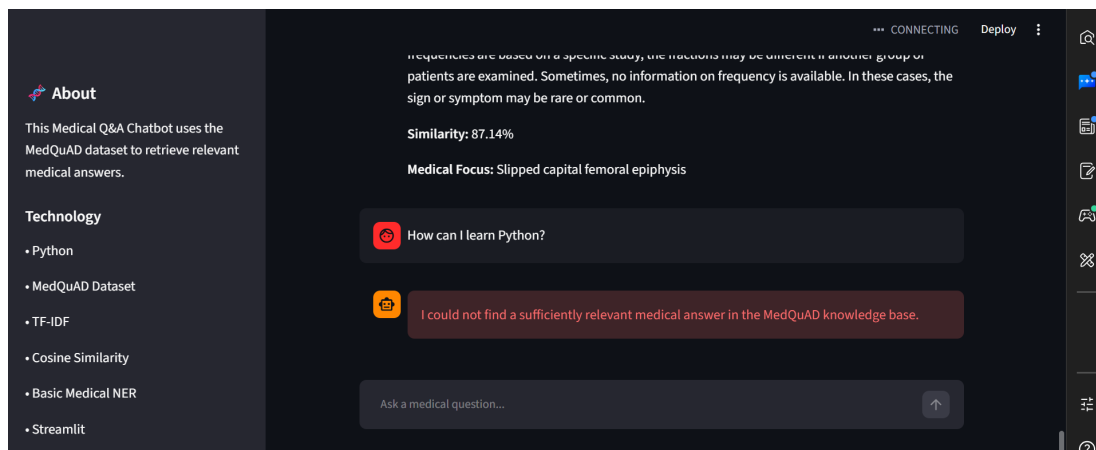


Figure 9. Non-medical question rejection demonstrating relevance-threshold behavior.

11. Safety, Limitations & Future Scope

Safety: The chatbot is a retrieval-based educational application. It does not diagnose conditions, prescribe medication, or replace a healthcare professional. The interface therefore includes a clear safety warning.

Limitations: Retrieval quality depends on the MedQuAD knowledge base and TF-IDF vocabulary. Paraphrased questions, spelling variations, or questions outside the dataset may receive low similarity. Basic entity recognition is also limited compared with advanced clinical NLP systems.

Future scope: Possible improvements include transformer-based embeddings, semantic search, FAISS/vector databases, improved medical NER models, multilingual support, conversation history, source citations, evaluation metrics, and stronger safety filtering.

12. Project Structure & Execution

A typical project structure is:

```
Medical_QA_Chatbot/
├── app.py
├── data / MedQuAD dataset files
├── retrieval / preprocessing components
├── requirements.txt
└── README.md
```

Typical local execution: create/activate a Python virtual environment, install the dependencies from requirements.txt, ensure the MedQuAD data is available, and start the Streamlit application using the project's Streamlit entry file.

13. Links & References

GitHub Repository: https://github.com/SujataMali03/Medical_QA_Chatbot

MedQuAD Dataset: <https://github.com/abachaa/MedQuAD>

The project report is based on the implemented chatbot, the supplied MedQuAD repository, and the application screenshots included as evidence in this document.

14. Conclusion

The Medical Q&A Chatbot demonstrates how a specialized knowledge base can be combined with information-retrieval techniques and a simple web interface to answer medical questions. The project meets the core requirements by using the MedQuAD dataset, implementing TF-IDF and cosine-similarity retrieval, providing basic medical entity recognition, and exposing the functionality through Streamlit. The included screenshots demonstrate the working interface, successful medical

retrieval examples, and safe handling of non-medical questions.

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